

Customer Churn Analysis

Project Overview

This project analyzes customer churn behavior using the Telco Customer Churn dataset. The objective is to identify key factors influencing customer attrition and provide actionable business insights that can help improve customer retention.

Customer churn is one of the most critical challenges for subscription-based businesses. Understanding why customers leave enables organizations to design targeted retention strategies and reduce revenue loss.

Dataset Information

The dataset contains customer demographic information, account details, subscription services, contract types, payment methods, tenure, and churn status.

Key Features

- Customer ID
 - Gender
 - Senior Citizen Status
 - Tenure
 - Contract Type
 - Payment Method
 - Internet Service
 - Online Security
 - Online Backup
 - Device Protection
 - Tech Support
 - Streaming Services
 - Monthly Charges
 - Total Charges
 - Churn Status
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Project Workflow

1. Data Cleaning

- Imported required Python libraries:
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Replaced blank values in **TotalCharges** with 0.
 - Converted **TotalCharges** to numeric format.
 - Checked for:
 - Missing values
 - Duplicate customer records
 - Converted **SeniorCitizen** values from:
 - 0 → No
 - 1 → Yes
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2. Exploratory Data Analysis (EDA)

The following analyses were performed:

Customer Churn Distribution

- Count of churned vs retained customers.
- Percentage breakdown of churned customers.

Demographic Analysis

- Churn by Gender
- Churn by Senior Citizen Status

Customer Tenure Analysis

- Distribution of customer tenure.
- Relationship between tenure and churn.

Contract Analysis

- Churn comparison across:
 - Month-to-Month Contracts
 - One-Year Contracts
 - Two-Year Contracts

Service-Based Analysis

Examined churn behavior across:

- Phone Service
- Multiple Lines
- Internet Service

- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies

Payment Method Analysis

- Churn patterns across different payment methods.
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Key Insights

1. Overall Churn Rate

- Approximately **26.5%** of customers have churned.
- The majority of customers remain active, but churn still represents a significant business concern.

2. Senior Citizens Are More Likely to Churn

- Senior citizens exhibit a noticeably higher churn rate than non-senior customers.
- This segment may require targeted retention programs and personalized support.

3. New Customers Are at Higher Risk

- Customers with short tenure (especially 1–2 months) show the highest churn rates.
- Long-term customers are significantly more likely to remain loyal.

4. Contract Type Strongly Influences Retention

- Month-to-Month customers have the highest churn rates.
- Customers with One-Year and Two-Year contracts are much more likely to stay.

5. Internet Services Impact Churn

- Fiber Optic users show higher churn rates compared to DSL users.
- Customers without internet services demonstrate lower churn behavior.

6. Support Services Reduce Churn

Customers using:

- Online Security
- Online Backup
- Device Protection

- Tech Support

generally show lower churn rates than customers without these services.

7. Payment Method Matters

- Customers using **Electronic Check** have the highest churn tendency.
 - Other payment methods demonstrate better customer retention.
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Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
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Business Impact

This analysis helps businesses:

- Identify customers at high risk of churn.
 - Improve onboarding strategies for new customers.
 - Promote long-term contract adoption.
 - Encourage adoption of support-related services.
 - Design targeted retention campaigns for vulnerable customer segments.
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Future Improvements

- Build a churn prediction model using Machine Learning.
 - Perform feature importance analysis.
 - Develop an interactive dashboard using Power BI or Tableau.
 - Implement customer segmentation for personalized retention strategies.
 - Create automated churn monitoring reports.
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